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Adaptive negative word line voltage

Abstract

A storage device comprises: a non-volatile memory including control circuitry and an array of memory cells formed using a set of word lines and a set of bit lines. A controller, coupled to the non-volatile memory, configured to: determine, based on a stage of a product lifetime of the non-volatile memory, a negative word line setting for implementing during performance of a first operation; perform the first operation, the first operation including adjusting, based on the negative word line setting, a negative word line relative parameter; determine, based on another stage of the product lifetime of the non-volatile memory, another negative word line setting for implementing during performance of a second operation; and perform the second operation, the second operation including adjusting, based on the other negative word line setting, another negative word line relative parameter.

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Background/Summary

FIELD

(1) This application relates to non-volatile memory apparatuses and the operation of non-volatile memory apparatuses.

BACKGROUND

(2) This section provides background information related to the technology associated with the present disclosure and, as such, is not necessarily prior art.

(3) Semiconductor memory apparatuses have become more popular for use in various electronic devices. For example, non-volatile semiconductor memory is used in cellular telephones, digital cameras, personal digital assistants, mobile computing devices, non-mobile computing devices and other devices.

(4) A charge-storing material such as a floating gate or a charge-trapping material can be used in such memory apparatuses to store a charge which represents a data state. A charge-trapping material can be arranged vertically in a three-dimensional (3D) stacked memory structure, or horizontally in a two-dimensional (2D) memory structure. One example of a 3D memory structure is the Bit Cost Scalable (BiCS) architecture which comprises a stack of alternating conductive and dielectric layers.

SUMMARY

(5) This section provides a general summary of the present disclosure and is not a comprehensive disclosure of its full scope or all of its features and advantages.

(6) An object of the present disclosure is to provide a memory apparatus and a method of operation of the memory apparatus that address and overcome shortcomings described herein.

(7) Accordingly, it is an aspect of the present disclosure to a storage device that comprises: a non-

volatile memory including control circuitry and an array of memory cells formed using a set of word lines and a set of bit lines; and a controller, coupled to the non-volatile memory. The controller is configured to: determine, based on a stage of a product lifetime of the non-volatile memory, a negative wordline setting for implementing during performance of a first operation; perform the first operation, the first operation including adjusting, based on the negative wordline setting, a negative wordline relative parameter; determine, based on another stage of the product lifetime of the non-volatile memory, another negative wordline setting for implementing during performance of a second operation; and perform the second operation, the second operation including adjusting, based on the other negative wordline setting, another negative wordline relative parameter.

(8) Accordingly, it is another aspect of the present disclosure to a method of operating a non-volatile semiconductor memory device. The method of determining, based on a stage of a product lifetime of the non-volatile memory, a negative wordline setting for implementing during performance of a first operation; performing the first operation, the first operation including adjusting, based on the negative wordline setting, a negative wordline relative parameter; determining, based on another stage of the product lifetime of the non-volatile memory, another negative wordline setting for implementing during performance of a second operation; and performing the second operation, the second operation including adjusting, based on the other negative wordline setting, another negative wordline relative parameter.

(9) Accordingly, it is another aspect of the present disclosure to an apparatus comprising: means for determining, based on a stage of a product lifetime of the non-volatile memory, a negative wordline setting for implementing during performance of a first operation; means for performing the first operation, the first operation including adjusting, based on the negative wordline setting, a negative wordline relative parameter; means for determining, based on another stage of the product lifetime of the non-volatile memory, another negative wordline setting for implementing during performance of a second operation; and means for performing the second operation, the second operation including adjusting, based on the other negative wordline setting, another negative wordline relative parameter.

(10) Further areas of applicability will become apparent from the description provided herein. The description and specific examples in this summary are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a detailed description of example embodiments, reference will now be made to the accompanying drawings in which:

(2) FIG. 1A is a block diagram of an example memory device;

(3) FIG. 1B is a block diagram of an example control circuit which comprises a programming circuit, a counting circuit, and a determination circuit;

(4) FIG. 2 depicts blocks of memory cells in an example two-dimensional configuration of the memory array of FIG. 1;

(5) FIG. 3A depicts a cross-sectional view of example floating gate memory cells in NAND strings;

(6) FIG. 3B depicts a cross-sectional view of the structure of FIG. 3A along line 329;

(7) FIG. 4A depicts a cross-sectional view of example charge-trapping memory cells in NAND strings;

(8) FIG. 4B depicts a cross-sectional view of the structure of FIG. 4A along line 429;

(9) FIG. 5A depicts an example block diagram of the sense block SB1 of FIG. 1;

(10) FIG. 5B depicts another example block diagram of the sense block SB1 of FIG. 1;

- (11) FIG. 6A is a perspective view of a set of blocks in an example three-dimensional configuration of the memory array of FIG. 1;
- (12) FIG. 6B depicts an example cross-sectional view of a portion of one of the blocks of FIG. 6A;
- (13) FIG. 6C depicts a plot of memory hole diameter in the stack of FIG. 6B;
- (14) FIG. 6D depicts a close-up view of the region 622 of the stack of FIG. 6B;
- (15) FIG. 7A depicts a top view of an example word line layer WLL0 of the stack of FIG. 6B;
- (16) FIG. 7B depicts a top view of an example top dielectric layer DL19 of the stack of FIG. 6B;
- (17) FIG. 8A depicts example NAND strings in the sub-blocks SBa-SBd of FIG. 7A;
- (18) FIG. 8B depicts another example view of NAND strings in sub-blocks;
- (19) FIG. 8C depicts a top view of example word line layers of a stack;
- (20) FIG. 9 depicts the V_{th} distributions of memory cells in an example one-pass programming operation with four data states;
- (21) FIG. 10 depicts the V_{th} distributions of memory cells in an example one-pass programming operation with eight data states;
- (22) FIG. 11 depicts the V_{th} distributions of memory cells in an example one-pass programming operation with sixteen data states;
- (23) FIG. 12 is a schematic voltage waveform for an example programming operation in a memory device;
- (24) FIGS. 13A and 13B depict the V_{th} distributions of memory cells;
- (25) FIGS. 14A and 14B provide example V_{th} distributions of a memory device array without and with application of a negative wordline, respectively;
- (26) FIGS. 15A and 15B provide an example of a static negative wordline scheme and an adaptive negative wordline scheme, respectively;
- (27) FIGS. 16A and 16B provide an example of a static negative wordline scheme and an adaptive negative wordline scheme, respectively;
- (28) FIG. 17 depicts an erase-program window oriented cell source voltage tuning for wordline, string, and EP-cycle variation compensation
- (29) FIG. 18 depicts a method of applying an adaptive negative wordline scheme, in accordance with embodiments described herein.

DETAILED DESCRIPTION

- (30) In the following description, details are set forth to provide an understanding of the present disclosure. In some instances, certain circuits, structures and techniques have not been described or shown in detail in order not to obscure the disclosure.
- (31) In general, the present disclosure relates to non-volatile memory apparatuses of the type well-suited for use in many applications. The non-volatile memory apparatus and associated methods of forming of this disclosure will be described in conjunction with one or more example embodiments. However, the specific example embodiments disclosed are merely provided to describe the inventive concepts, features, advantages and objectives with sufficient clarity to permit those skilled in this art to understand and practice the disclosure. Specifically, the example embodiments are provided so that this disclosure will be thorough, and will fully convey the scope to those who are skilled in the art. Numerous specific details are set forth such as examples of specific components, devices, and methods, to provide a thorough understanding of embodiments of the present disclosure. It will be apparent to those skilled in the art that specific details need not be employed, that example embodiments may be embodied in many different forms and that neither should be construed to limit the scope of the disclosure. In some example embodiments, well-known processes, well-known device structures, and well-known technologies are not described in detail.
- (32) Various terms are used to refer to particular system components. Different companies may refer to a component by different names—this document does not intend to distinguish between components that differ in name but not function. In the following discussion and in the claims, the

terms “including” and “comprising” are used in an open-ended fashion, and thus should be interpreted to mean “including, but not limited to” Also, the term “couple” or “couples” is intended to mean either an indirect or direct connection. Thus, if a first device couples to a second device, that connection may be through a direct connection or through an indirect connection via other devices and connections.

(33) Additionally, when a layer or element is referred to as being “on” another layer or substrate, it can be directly on the other layer of substrate, or intervening layers may also be present. Further, it will be understood that when a layer is referred to as being “under” another layer, it can be directly under, and one or more intervening layers may also be present. Furthermore, when a layer is referred to as “between” two layers, it can be the only layer between the two layers, or one or more intervening layers may also be present.

(34) As described, non-volatile memory systems are a type of memory that retains stored information without requiring an external power source. Non-volatile memory is widely used in various electronic devices and in stand-alone memory devices. For example, non-volatile memory can be found in laptops, digital audio player, digital cameras, smart phones, video games, scientific instruments, industrial robots, medical electronics, solid-state drives, USB drives, memory cards, and the like. Non-volatile memory can be electronically programmed/reprogrammed and erased.

(35) Examples of non-volatile memory systems include flash memory, such as NAND flash or NOR flash. NAND flash memory structures typically arrange multiple memory cell transistors (e.g., floating-gate transistors or charge trap transistors) in series with and between two select gates (e.g., a drain-side select gate and a source-side select gate). The memory cell transistors in series and the select gates may be referred to as a NAND string. NAND flash memory may be scaled in order to reduce cost per bit.

(36) Conventionally, programmed state levels are higher than a V_{th} value of 0V because a V_{th} value below 0V cannot be sensed or verified. The V_{th} distribution without negative word line (NWL) depicts higher states (which are further from neutral V_{th} than lower states) having large vertical data retention (DR) degradation. This issue is worse for the end-of-life (EOL) stage of the memory device, where the vertical DR degradation is much larger due to the wear-out of the tunneling oxide. NWL is a scheme used in reducing DR degradation. The V_{th} distribution with NWL depicts implementing NWL scheme, where the erase/program window and all states V_{th} can be shifted down, thereby allowing the higher states to be closer to the neutral V_{th} level (i.e., no charge in the floating gate). As a result, the vertical DR degradation may be improved.

(37) Due to aggressive scaling and cost reduction, there is new process improvement, such as smaller ON pitch, Y-size shrink, and MANOS improvement. After introducing NWL, many process changes can significantly impact the DR degradation for begin-of-life (BOL) stage of the product lifetime of the memory device. However, not implementing NWL would lose the benefit for the EOL stage for a memory device. To resolve this, embodiments describe herein are directed to an adaptive NWL scheme which helps reduce DR degradation for the BOL stage but simultaneously maintaining NWL benefit for the EOL stage.

(38) To help further illustrate the foregoing, FIG. 1A will now be described. FIG. 1A is a block diagram of an example memory device. The memory device **100** may include one or more memory die **108**. The memory die **108** includes a memory structure **126** of memory cells, such as an array of memory cells, control circuitry **110**, and read/write circuits **128**. The memory structure **126** is addressable by word lines via a row decoder **124** and by bit lines via a column decoder **132**. The read/write circuits **128** include multiple sense blocks SB1, SB2, . . . , SBp (sensing circuitry) and allow a page of memory cells to be read or programmed in parallel. Typically, a controller **122** is included in the same memory device **100** (e.g., a removable storage card) as the one or more memory die **108**. Commands and data are transferred between the host **140** and controller **122** via a data bus **120**, and between the controller and the one or more memory die **108** via lines **118**.

(39) The memory structure can be 2D or 3D. The memory structure may comprise one or more

array of memory cells including a 3D array. The memory structure may comprise a monolithic three dimensional memory structure in which multiple memory levels are formed above (and not in) a single substrate, such as a wafer, with no intervening substrates. The memory structure may comprise any type of non-volatile memory that is monolithically formed in one or more physical levels of arrays of memory cells having an active area disposed above a silicon substrate. The memory structure may be in a non-volatile memory device having circuitry associated with the operation of the memory cells, whether the associated circuitry is above or within the substrate.

(40) The control circuitry **110** cooperates with the read/write circuits **128** to perform memory operations on the memory structure **126**, and includes a state machine **112**, an on-chip address decoder **114**, and a power control module **116**. The state machine **112** provides chip-level control of memory operations. A storage region **113** may be provided, e.g., for verify parameters as described herein.

(41) The on-chip address decoder **114** provides an address interface between that used by the host or a memory controller to the hardware address used by the decoders **124** and **132**. The power control module **116** controls the power and voltages supplied to the word lines and bit lines during memory operations. It can include drivers for word lines, SGS and SGD transistors and source lines. The sense blocks can include bit line drivers, in one approach. An SGS transistor is a select gate transistor at a source end of a NAND string, and an SGD transistor is a select gate transistor at a drain end of a NAND string.

(42) In some implementations, some of the components can be combined. In various designs, one or more of the components (alone or in combination), other than memory structure **126**, can be thought of as at least one control circuit which is configured to perform the actions described herein. For example, a control circuit may include any one of, or a combination of, control circuitry **110**, state machine **112**, decoders **114/132**, power control module **116**, sense blocks SBb, SB2, . . . , SBp, read/write circuits **128**, controller **122**, and so forth.

(43) The control circuits can include a programming circuit configured to program memory cells of a word line of a block and verify the set of the memory cells. The control circuits can also include a counting circuit configured to determine a number of memory cells that are verified to be in a data state. The control circuits can also include a determination circuit configured to determine, based on the number, whether the block is faulty.

(44) For example, FIG. **1B** is a block diagram of an example control circuit **150** which comprises a programming circuit **151**, a counting circuit **152** and a determination circuit **153**. The programming circuit may include software, firmware and/or hardware. The counting circuit may include software, firmware and/or hardware which implements. The determination circuit may include software, firmware and/or hardware which implements.

(45) The off-chip controller **122** may comprise a processor **122c**, storage devices (memory) such as ROM **122a** and RAM **122b** and an error-correction code (ECC) engine **245**. The ECC engine can correct a number of read errors which are caused when the upper tail of a V_{th} distribution becomes too high. However, uncorrectable errors may exist in some cases. The techniques provided herein reduce the likelihood of uncorrectable errors.

(46) The storage device comprises code such as a set of instructions, and the processor is operable to execute the set of instructions to provide the functionality described herein. Alternatively or additionally, the processor can access code from a storage device **126a** of the memory structure, such as a reserved area of memory cells in one or more word lines.

(47) For example, code can be used by the controller **122** to access the memory structure such as for programming, read and erase operations. The code can include boot code and control code (e.g., set of instructions). The boot code is software that initializes the controller during a booting or startup process and enables the controller to access the memory structure. The code can be used by the controller to control one or more memory structures. Upon being powered up, the processor **122c** fetches the boot code from the ROM **122a** or storage device **126a** for execution, and the boot

code initializes the system components and loads the control code into the RAM 122b. Once the control code is loaded into the RAM, it is executed by the processor. The control code includes drivers to perform basic tasks such as controlling and allocating memory, prioritizing the processing of instructions, and controlling input and output ports.

(48) In one embodiment, the host is a computing device (e.g., laptop, desktop, smartphone, tablet, digital camera) that includes one or more processors, one or more processor readable storage devices (RAM, ROM, flash memory, hard disk drive, solid state memory) that store processor readable code (e.g., software) for programming the one or more processors to perform the methods described herein. The host may also include additional system memory, one or more input/output interfaces and/or one or more input/output devices in communication with the one or more processors.

(49) Other types of non-volatile memory in addition to NAND flash memory can also be used.

(50) Semiconductor memory devices include volatile memory devices, such as dynamic random access memory (“DRAM”) or static random access memory (“SRAM”) devices, non-volatile memory devices, such as resistive random access memory (“ReRAM”), electrically erasable programmable read only memory (“EEPROM”), flash memory (which can also be considered a subset of EEPROM), ferroelectric random access memory (“FRAM”), and magnetoresistive random access memory (“MRAM”), and other semiconductor elements capable of storing information. Each type of memory device may have different configurations. For example, flash memory devices may be configured in a NAND or a NOR configuration.

(51) The smallest piece of a NAND flash die is a cell, and each cell is stored in a page. Each page can be written to, and they are the smallest piece of the NAND flash that can store data or be programmed. Groups of pages are called blocks. There may be over 100 pages in each block. Because multiple pages are contained in each block, blocks can store a large amount of data. When it is necessary to erase part of the data stored in the NAND flash memory, it can only be erased by block. It is not possible to erase smaller or larger groups of data within a NAND flash die.

(52) When blocks are grouped together, they form planes. Planes then form NAND flash dies. Dies can contain a single plane full of data blocks, or they may feature multiple planes that have been linked together. The number and configurations of planes within the NAND flash die is adaptable.

(53) Further, the memory devices can be formed from passive and/or active elements, in any combinations. By way of non-limiting example, passive semiconductor memory elements include ReRAM device elements, which in some embodiments include a resistivity switching storage element, such as an anti-fuse or phase change material, and optionally a steering element, such as a diode or transistor. Further by way of non-limiting example, active semiconductor memory elements include EEPROM and flash memory device elements, which in some embodiments include elements containing a charge storage region, such as a floating gate, conductive nanoparticles, or a charge storage dielectric material.

(54) Multiple memory elements may be configured so that they are connected in series or so that each element is individually accessible. By way of non-limiting example, flash memory devices in a NAND configuration (NAND memory) typically contain memory elements connected in series. A NAND string is an example of a set of series-connected transistors comprising memory cells and SG transistors.

(55) A NAND memory array may be configured so that the array is composed of multiple strings of memory in which a string is composed of multiple memory elements sharing a single bit line and accessed as a group. Alternatively, memory elements may be configured so that each element is individually accessible, e.g., a NOR memory array. NAND and NOR memory configurations are examples, and memory elements may be otherwise configured.

(56) The semiconductor memory elements located within and/or over a substrate may be arranged in two or three dimensions, such as a two dimensional memory structure or a three dimensional memory structure.

(57) In a two dimensional memory structure, the semiconductor memory elements are arranged in a single plane or a single memory device level. Typically, in a two dimensional memory structure, memory elements are arranged in a plane (e.g., in an x-y direction plane) which extends substantially parallel to a major surface of a substrate that supports the memory elements. The substrate may be a wafer over or in which the layer of the memory elements are formed or it may be a carrier substrate which is attached to the memory elements after they are formed. As a non-limiting example, the substrate may include a semiconductor such as silicon.

(58) The memory elements may be arranged in the single memory device level in an ordered array, such as in a plurality of rows and/or columns. However, the memory elements may be arrayed in non-regular or non-orthogonal configurations. The memory elements may each have two or more electrodes or contact lines, such as bit lines and word lines.

(59) A three dimensional memory array is arranged so that memory elements occupy multiple planes or multiple memory device levels, thereby forming a structure in three dimensions (i.e., in the x, y and z directions, where the z direction is substantially perpendicular and the x and y directions are substantially parallel to the major surface of the substrate).

(60) As a non-limiting example, a three dimensional memory structure may be vertically arranged as a stack of multiple two dimensional memory device levels. As another non-limiting example, a three dimensional memory array may be arranged as multiple vertical columns (e.g., columns extending substantially perpendicular to the major surface of the substrate, i.e., in the y direction) with each column having multiple memory elements. The columns may be arranged in a two dimensional configuration, e.g., in an x-y plane, resulting in a three dimensional arrangement of memory elements with elements on multiple vertically stacked memory planes. Other configurations of memory elements in three dimensions can also constitute a three dimensional memory array.

(61) By way of non-limiting example, in a three dimensional NAND memory array, the memory elements may be coupled together to form a NAND string within a single horizontal (e.g., x-y) memory device level. Alternatively, the memory elements may be coupled together to form a vertical NAND string that traverses across multiple horizontal memory device levels. Other three dimensional configurations can be envisioned wherein some NAND strings contain memory elements in a single memory level while other strings contain memory elements which span through multiple memory levels. Three dimensional memory arrays may also be designed in a NOR configuration and in a ReRAM configuration.

(62) Typically, in a monolithic three dimensional memory array, one or more memory device levels are formed above a single substrate. Optionally, the monolithic three dimensional memory array may also have one or more memory layers at least partially within the single substrate. As a non-limiting example, the substrate may include a semiconductor such as silicon. In a monolithic three dimensional array, the layers constituting each memory device level of the array are typically formed on the layers of the underlying memory device levels of the array. However, layers of adjacent memory device levels of a monolithic three dimensional memory array may be shared or have intervening layers between memory device levels.

(63) Then again, two dimensional arrays may be formed separately and then packaged together to form a non-monolithic memory device having multiple layers of memory. For example, non-monolithic stacked memories can be constructed by forming memory levels on separate substrates and then stacking the memory levels atop each other. The substrates may be thinned or removed from the memory device levels before stacking, but as the memory device levels are initially formed over separate substrates, the resulting memory arrays are not monolithic three dimensional memory arrays. Further, multiple two dimensional memory arrays or three dimensional memory arrays (monolithic or non-monolithic) may be formed on separate chips and then packaged together to form a stacked-chip memory device.

(64) Associated circuitry is typically required for operation of the memory elements and for

communication with the memory elements. As non-limiting examples, memory devices may have circuitry used for controlling and driving memory elements to accomplish functions such as programming and reading. This associated circuitry may be on the same substrate as the memory elements and/or on a separate substrate. For example, a controller for memory read-write operations may be located on a separate controller chip and/or on the same substrate as the memory elements.

(65) One of skill in the art will recognize that this technology is not limited to the two dimensional and three dimensional exemplary structures described but covers all relevant memory structures within the spirit and scope of the technology as described herein and as understood by one of skill in the art.

(66) FIG. 2 depicts blocks of memory cells in an example two-dimensional configuration of the memory array **126** of FIG. 1. The memory array can include many blocks. Each example block **200**, **210** includes a number of NAND strings and respective bit lines, e.g., **BL0**, **BL1**, . . . which are shared among the blocks. Each NAND string is connected at one end to a drain select gate (SGD), and the control gates of the drain select gates are connected via a common SGD line. The NAND strings are connected at their other end to a source select gate which, in turn, is connected to a common source line **220**. Sixteen word lines, for example, **WL0-WL15**, extend between the source select gates and the drain select gates. In some cases, dummy word lines, which contain no user data, can also be used in the memory array adjacent to the select gate transistors. Such dummy word lines can shield the edge data word line from certain edge effects.

(67) One type of non-volatile memory which may be provided in the memory array is a floating gate memory. See FIGS. 3A and 3B. Other types of non-volatile memory can also be used. For example, a charge-trapping memory cell uses a non-conductive dielectric material in place of a conductive floating gate to store charge in a non-volatile manner. See FIGS. 4A and 4B. A triple layer dielectric formed of silicon oxide, silicon nitride and silicon oxide (“ONO”) is sandwiched between a conductive control gate and a surface of a semi-conductive substrate above the memory cell channel. The cell is programmed by injecting electrons from the cell channel into the nitride, where they are trapped and stored in a limited region. This stored charge then changes the threshold voltage of a portion of the channel of the cell in a manner that is detectable. The cell is erased by injecting hot holes into the nitride. A similar cell can be provided in a split-gate configuration where a doped polysilicon gate extends over a portion of the memory cell channel to form a separate select transistor.

(68) In another approach, NROM cells are used. Two bits, for example, are stored in each NROM cell, where an ONO dielectric layer extends across the channel between source and drain diffusions. The charge for one data bit is localized in the dielectric layer adjacent to the drain, and the charge for the other data bit localized in the dielectric layer adjacent to the source. Multi-state data storage is obtained by separately reading binary states of the spatially separated charge storage regions within the dielectric. Other types of non-volatile memory are also known.

(69) FIG. 3A depicts a cross-sectional view of example floating gate memory cells in NAND strings. A bit line or NAND string direction goes into the page, and a word line direction goes from left to right. As an example, word line **324** extends across NAND strings which include respective channel regions **306**, **316** and **326**. The memory cell **300** includes a control gate **302**, a floating gate **304**, a tunnel oxide layer **305** and the channel region **306**. The memory cell **310** includes a control gate **312**, a floating gate **314**, a tunnel oxide layer **315** and the channel region **316**. The memory cell **320** includes a control gate **322**, a floating gate **321**, a tunnel oxide layer **325** and the channel region **326**. Each memory cell is in a different respective NAND string. An inter-poly dielectric (IPD) layer **328** is also depicted. The control gates are portions of the word line. A cross-sectional view along line **329** is provided in FIG. 3B.

(70) The control gate wraps around the floating gate, increasing the surface contact area between the control gate and floating gate. This results in higher IPD capacitance, leading to a higher

coupling ratio which makes programming and erase easier. However, as NAND memory devices are scaled down, the spacing between neighboring cells becomes smaller so there is almost no space for the control gate and the IPD between two adjacent floating gates. As an alternative, as shown in FIGS. 4A and 4B, the flat or planar memory cell has been developed in which the control gate is flat or planar; that is, it does not wrap around the floating gate, and its only contact with the charge storage layer is from above it. In this case, there is no advantage in having a tall floating gate. Instead, the floating gate is made much thinner. Further, the floating gate can be used to store charge, or a thin charge trap layer can be used to trap charge. This approach can avoid the issue of ballistic electron transport, where an electron can travel through the floating gate after tunneling through the tunnel oxide during programming.

(71) FIG. 3B depicts a cross-sectional view of the structure of FIG. 3A along line 329. The NAND string 330 includes an SGS transistor 331, example memory cells 300, 333, . . . , 334 and 335, and an SGD transistor 336. The memory cell 300, as an example of each memory cell, includes the control gate 302, the IPD layer 328, the floating gate 304 and the tunnel oxide layer 305, consistent with FIG. 3A. Passageways in the IPD layer in the SGS and SGD transistors allow the control gate layers and floating gate layers to communicate. The control gate and floating gate layers may be polysilicon and the tunnel oxide layer may be silicon oxide, for instance. The IPD layer can be a stack of nitrides (N) and oxides (O) such as in a N—O—N—O—N configuration.

(72) The NAND string may be formed on a substrate which comprises a p-type substrate region 355, an n-type well 356 and a p-type well 357. N-type source/drain diffusion regions sd1, sd2, sd3, sd4, sd5, sd6 and sd7 are formed in the p-type well. A channel voltage, V_{ch} , may be applied directly to the channel region of the substrate.

(73) FIG. 4A depicts a cross-sectional view of example charge-trapping memory cells in NAND strings. The view is in a word line direction of memory cells comprising a flat control gate and charge-trapping regions as a 2D example of memory cells in the memory cell array 126 of FIG. 1. Charge-trapping memory can be used in NOR and NAND flash memory device. This technology uses an insulator such as a SiN film to store electrons, in contrast to a floating-gate MOSFET technology which uses a conductor such as doped polycrystalline silicon to store electrons. As an example, a word line (WL) 424 extends across NAND strings which include respective channel regions 406, 416 and 426. Portions of the word line provide control gates 402, 412 and 422. Below the word line is an IPD layer 428, charge-trapping layers 404, 414 and 421, polysilicon layers 405, 415 and 425 and tunneling layer layers 409, 407 and 408. Each charge-trapping layer extends continuously in a respective NAND string.

(74) A memory cell 400 includes the control gate 402, the charge-trapping layer 404, the polysilicon layer 405 and a portion of the channel region 406. A memory cell 410 includes the control gate 412, the charge-trapping layer 414, a polysilicon layer 415 and a portion of the channel region 416. A memory cell 420 includes the control gate 422, the charge-trapping layer 421, the polysilicon layer 425 and a portion of the channel region 426.

(75) A flat control gate is used here instead of a control gate that wraps around a floating gate. One advantage is that the charge-trapping layer can be made thinner than a floating gate. Additionally, the memory cells can be placed closer together.

(76) FIG. 4B depicts a cross-sectional view of the structure of FIG. 4A along line 429. The view shows a NAND string 430 having a flat control gate and a charge-trapping layer. The NAND string 430 includes an SGS transistor 431, example memory cells 400, 433, . . . , 434 and 435, and an SGD transistor 436.

(77) The NAND string may be formed on a substrate which comprises a p-type substrate region 455, an n-type well 456 and a p-type well 457. N-type source/drain diffusion regions sd1, sd2, sd3, sd4, sd5, sd6 and sd7 are formed in the p-type well 457. A channel voltage, V_{ch} , may be applied directly to the channel region of the substrate. The memory cell 400 includes the control gate 402 and the IPD layer 428 above the charge-trapping layer 404, the polysilicon layer 405, the tunneling

layer **409** and the channel region **406**.

(78) The control gate layer may be polysilicon and the tunneling layer may be silicon oxide, for instance. The IPD layer can be a stack of high-k dielectrics such as AlO_x or HfO_x which help increase the coupling ratio between the control gate layer and the charge-trapping or charge storing layer. The charge-trapping layer can be a mix of silicon nitride and oxide, for instance.

(79) The SGD and SGS transistors have the same configuration as the memory cells but with a longer channel length to ensure that current is cutoff in an inhibited NAND string.

(80) In this example, the layers **404**, **405** and **409** extend continuously in the NAND string. In another approach, portions of the layers **404**, **405** and **409** which are between the control gates **402**, **412** and **422** can be removed, exposing a top surface of the channel **406**.

(81) FIG. 5A depicts an example block diagram of the sense block SB1 of FIG. 1. In one approach, a sense block comprises multiple sense circuits. Each sense circuit is associated with data latches. For example, the example sense circuits **550a**, **551a**, **552a** and **553a** are associated with the data latches **550b**, **551b**, **552b** and **553b**, respectively. In one approach, different subsets of bit lines can be sensed using different respective sense blocks. This allows the processing load which is associated with the sense circuits to be divided up and handled by a respective processor in each sense block. For example, a sense circuit controller **560** in SB1 can communicate with the set of sense circuits and latches. The sense circuit controller may include a pre-charge circuit **561** which provides a voltage to each sense circuit for setting a pre-charge voltage. In one possible approach, the voltage is provided to each sense circuit independently, e.g., via the data bus, DBUS **503** and a local bus such as LBUS1 or LBUS2 in FIG. 5B. In another possible approach, a common voltage is provided to each sense circuit concurrently, e.g., via the line **505** in FIG. 5B. The sense circuit controller may also include a memory **562** and a processor **563**. As mentioned also in connection with FIG. 2, the memory **562** may store code which is executable by the processor to perform the functions described herein. These functions can include reading latches which are associated with the sense circuits, setting bit values in the latches and providing voltages for setting pre-charge levels in sense nodes of the sense circuits. Further example details of the sense circuit controller and the sense circuits **550a** and **551a** are provided below.

(82) FIG. 5B depicts another example block diagram of the sense block SB1 of FIG. 1. The sense circuit controller **560** communicates with multiple sense circuits including example sense circuits **550a** and **551a**, also shown in FIG. 5A. The sense circuit **550a** includes latches **550b**, including a trip latch **526**, an offset verify latch **527** and data state latches **528**. The sense circuit further includes a voltage clamp **521** such as a transistor which sets a pre-charge voltage at a sense node **522**. A sense node to bit line (BL) switch **523** selectively allows the sense node to communicate with a bit line **525**, e.g., the sense node is electrically connected to the bit line so that the sense node voltage can decay. The bit line **525** is connected to one or more memory cells such as a memory cell MC1. A voltage clamp **524** can set a voltage on the bit line, such as during a sensing operation or during a program voltage. A local bus, LBUS1, allows the sense circuit controller to communicate with components in the sense circuit, such as the latches **550b** and the voltage clamp in some cases. To communicate with the sense circuit **550a**, the sense circuit controller provides a voltage via a line **502** to a transistor **504** to connect LBUS1 with a data bus DBUS, **503**. The communicating can include sending data to the sense circuit and/or receive data from the sense circuit.

(83) The sense circuit controller can communicate with different sense circuits in a time-multiplexed manner, for instance. A line **505** may be connected to the voltage clamp in each sense circuit, in one approach.

(84) The sense circuit **551a** includes latches **551b**, including a trip latch **546**, an offset verify latch **547** and data state latches **548**. A voltage clamp **541** may be used to set a pre-charge voltage at a sense node **542**. A sense node to bit line (BL) switch **543** selectively allows the sense node to communicate with a bit line **545**, and a voltage clamp **544** can set a voltage on the bit line. The bit

line **545** is connected to one or more memory cells such as a memory cell **MC2**. A local bus, **LBUS2**, allows the sense circuit controller to communicate with components in the sense circuit, such as the latches **551b** and the voltage clamp in some cases. To communicate with the sense circuit **551a**, the sense circuit controller provides a voltage via a line **501** to a transistor **506** to connect **LBUS2** with **DBUS**.

(85) The sense circuit **550a** may be a first sense circuit which comprises a first trip latch **526** and the sense circuit **551a** may be a second sense circuit which comprises a second trip latch **546**.

(86) The sense circuit **550a** is an example of a first sense circuit comprising a first sense node **522**, where the first sense circuit is associated with a first memory cell **MC1** and a first bit line **525**. The sense circuit **551a** is an example of a second sense circuit comprising a second sense node **542**, where the second sense circuit is associated with a second memory cell **MC2** and a second bit line **545**.

(87) FIG. **6A** is a perspective view of a set of blocks **600** in an example three-dimensional configuration of the memory array **126** of FIG. **1**. On the substrate are example blocks **BLK0**, **BLK1**, **BLK2** and **BLK3** of memory cells (storage elements) and a peripheral area **604** with circuitry for use by the blocks. For example, the circuitry can include voltage drivers **605** which can be connected to control gate layers of the blocks. In one approach, control gate layers at a common height in the blocks are commonly driven. The substrate **601** can also carry circuitry under the blocks, along with one or more lower metal layers which are patterned in conductive paths to carry signals of the circuitry. The blocks are formed in an intermediate region **602** of the memory device. In an upper region **603** of the memory device, one or more upper metal layers are patterned in conductive paths to carry signals of the circuitry. Each block comprises a stacked area of memory cells, where alternating levels of the stack represent word lines. In one possible approach, each block has opposing tiered sides from which vertical contacts extend upward to an upper metal layer to form connections to conductive paths. While four blocks are depicted as an example, two or more blocks can be used, extending in the x- and/or y-directions.

(88) In one possible approach, the length of the plane, in the x-direction, represents a direction in which signal paths to word lines extend in the one or more upper metal layers (a word line or SGD line direction), and the width of the plane, in the y-direction, represents a direction in which signal paths to bit lines extend in the one or more upper metal layers (a bit line direction). The z-direction represents a height of the memory device.

(89) FIG. **6B** depicts an example cross-sectional view of a portion of one of the blocks of FIG. **6A**. The block comprises a stack **610** of alternating conductive and dielectric layers. In this example, the conductive layers comprise two SGD layers, two SGS layers and four dummy word line layers **DWLD0**, **DWLD1**, **DWLS0** and **DWLS1**, in addition to data word line layers (word lines) **WLL0-WLL10**. The dielectric layers are labelled as **DL0-DL19**. Further, regions of the stack which comprise NAND strings **NS1** and **NS2** are depicted. Each NAND string encompasses a memory hole **618** or **619** which is filled with materials which form memory cells adjacent to the word lines. A region **622** of the stack is shown in greater detail in FIG. **6D**.

(90) The stack includes a substrate **611**, an insulating film **612** on the substrate, and a portion of a source line **SL**. **NS1** has a source-end **613** at a bottom **614** of the stack and a drain-end **615** at a top **616** of the stack. Metal-filled slits **617** and **620** may be provided periodically across the stack as interconnects which extend through the stack, such as to connect the source line to a line above the stack. The slits may be used during the formation of the word lines and subsequently filled with metal. A portion of a bit line **BL0** is also depicted. A conductive via **621** connects the drain-end **615** to **BL0**.

(91) FIG. **6C** depicts a plot of memory hole diameter in the stack of FIG. **6B**. The vertical axis is aligned with the stack of FIG. **6B** and depicts a width (**wMH**), e.g., diameter, of the memory holes **618** and **619**. The word line layers **WLL0-WLL10** of FIG. **6A** are repeated as an example and are at respective heights **z0-z10** in the stack. In such a memory device, the memory holes which are

etched through the stack have a very high aspect ratio. For example, a depth-to-diameter ratio of about 25-30 is common. The memory holes may have a circular cross-section. Due to the etching process, the memory hole width can vary along the length of the hole. Typically, the diameter becomes progressively smaller from the top to the bottom of the memory hole. That is, the memory holes are tapered, narrowing at the bottom of the stack. In some cases, a slight narrowing occurs at the top of the hole near the select gate so that the diameter becomes slight wider before becoming progressively smaller from the top to the bottom of the memory hole.

(92) Due to the non-uniformity in the width of the memory hole, the programming speed, including the program slope and erase speed of the memory cells can vary based on their position along the memory hole, e.g., based on their height in the stack. With a smaller diameter memory hole, the electric field across the tunnel oxide is relatively stronger, so that the programming and erase speed is relatively higher. One approach is to define groups of adjacent word lines for which the memory hole diameter is similar, e.g., within a defined range of diameter, and to apply an optimized verify scheme for each word line in a group. Different groups can have different optimized verify schemes.

(93) FIG. 6D depicts a close-up view of the region 622 of the stack of FIG. 6B. Memory cells are formed at the different levels of the stack at the intersection of a word line layer and a memory hole. In this example, SGD transistors 680 and 681 are provided above dummy memory cells 682 and 683 and a data memory cell MC. A number of layers can be deposited along the sidewall (SW) of the memory hole 630 and/or within each word line layer, e.g., using atomic layer deposition. For example, each column (e.g., the pillar which is formed by the materials within a memory hole) can include a charge-trapping layer or film 663 such as SiN or other nitride, a tunneling layer 664, a polysilicon body or channel 665, and a dielectric core 666. A word line layer can include a blocking oxide/block high-k material 660, a metal barrier 661, and a conductive metal 662 such as Tungsten as a control gate. For example, control gates 690, 691, 692, 693 and 694 are provided. In this example, all of the layers except the metal are provided in the memory hole. In other approaches, some of the layers can be in the control gate layer. Additional pillars are similarly formed in the different memory holes. A pillar can form a columnar active area (AA) of a NAND string.

(94) When a memory cell is programmed, electrons are stored in a portion of the charge-trapping layer which is associated with the memory cell. These electrons are drawn into the charge-trapping layer from the channel, and through the tunneling layer. The V_{th} of a memory cell is increased in proportion to the amount of stored charge. During an erase operation, the electrons return to the channel.

(95) Each of the memory holes can be filled with a plurality of annular layers comprising a blocking oxide layer, a charge trapping layer, a tunneling layer and a channel layer. A core region of each of the memory holes is filled with a body material, and the plurality of annular layers are between the core region and the word line in each of the memory holes.

(96) The NAND string can be considered to have a floating body channel because the length of the channel is not formed on a substrate. Further, the NAND string is provided by a plurality of word line layers above one another in a stack, and separated from one another by dielectric layers.

(97) FIG. 7A depicts a top view of an example word line layer WLL0 of the stack of FIG. 6B. As mentioned, a 3D memory device can comprise a stack of alternating conductive and dielectric layers. The conductive layers provide the control gates of the SG transistors and memory cells. The layers used for the SG transistors are SG layers and the layers used for the memory cells are word line layers. Further, memory holes are formed in the stack and filled with a charge-trapping material and a channel material. As a result, a vertical NAND string is formed. Source lines are connected to the NAND strings below the stack and bit lines are connected to the NAND strings above the stack.

(98) A block BLK in a 3D memory device can be divided into sub-blocks, where each sub-block comprises a set of NAND string which have a common SGD control line. For example, see the

SGD lines/control gates SGD0, SGD1, SGD2 and SGD3 in the sub-blocks SBa, SBb, SBc and SBd, respectively. The sub-blocks SBa, SBb, SBc and SBd may also be referred herein as a string of memory cells of a word line. As described, a string of memory cells of a word line may include a plurality of memory cells that are part of the same sub-block, and that are also disposed in the same word line layer and/or that are configured to have their control gates biased by the same word line and/or with the same word line voltage.

(99) Further, a word line layer in a block can be divided into regions. Each region in a respective sub-block can extend between slits which are formed periodically in the stack to process the word line layers during the fabrication process of the memory device. This processing can include replacing a sacrificial material of the word line layers with metal. Generally, the distance between slits should be relatively small to account for a limit in the distance that an etchant can travel laterally to remove the sacrificial material, and that the metal can travel to fill a void which is created by the removal of the sacrificial material. For example, the distance between slits may allow for a few rows of memory holes between adjacent slits. The layout of the memory holes and slits should also account for a limit in the number of bit lines which can extend across the region while each bit line is connected to a different memory cell. After processing the word line layers, the slits can optionally be filled with metal to provide an interconnect through the stack.

(100) This figure and other are not necessarily to scale. In practice, the regions can be much longer in the x-direction relative to the y-direction than is depicted to accommodate additional memory holes.

(101) In this example, there are four rows of memory holes between adjacent slits. A row here is a group of memory holes which are aligned in the x-direction. Moreover, the rows of memory holes are in a staggered pattern to increase the density of the memory holes. The word line layer or word line is divided into regions WLL0 a, WLL0 b, WLL0 c and WLL0 d which are each connected by a connector 713. The last region of a word line layer in a block can be connected to a first region of a word line layer in a next block, in one approach. The connector, in turn, is connected to a voltage driver for the word line layer. The region WLL0 a has example memory holes 710 and 711 along a line 712. The region WLL0 b has example memory holes 714 and 715. The region WLL0 c has example memory holes 716 and 717. The region WLL0 d has example memory holes 718 and 719. The memory holes are also shown in FIG. 7B. Each memory hole can be part of a respective NAND string. For example, the memory holes 710, 714, 716 and 718 can be part of NAND strings NS0_SBa, NS0_SBb, NS0_SBc and NS0_SBd, respectively.

(102) Each circle represents the cross-section of a memory hole at a word line layer or SG layer. Example circles shown with dashed lines represent memory cells which are provided by the materials in the memory hole and by the adjacent word line layer. For example, memory cells 720 and 721 are in WLL0 a, memory cells 724 and 725 are in WLL0 b, memory cells 726 and 727 are in WLL0 c, and memory cells 728 and 729 are in WLL0 d. These memory cells are at a common height in the stack.

(103) Metal-filled slits 701, 702, 703 and 704 (e.g., metal interconnects) may be located between and adjacent to the edges of the regions WLL0 a-WLL0 d. The metal-filled slits provide a conductive path from the bottom of the stack to the top of the stack. For example, a source line at the bottom of the stack may be connected to a conductive line above the stack, where the conductive line is connected to a voltage driver in a peripheral region of the memory device. See also FIG. 8A for further details of the sub-blocks SBa-SBd of FIG. 7A.

(104) FIG. 7B depicts a top view of an example top dielectric layer DL19 of the stack of FIG. 6B. The dielectric layer is divided into regions DL19 a, DL19 b, DL19 c and DL19 d. Each region can be connected to a respective voltage driver. This allows a set of memory cells in one region of a word line layer to be programmed concurrently, with each memory cell being in a respective NAND string which is connected to a respective bit line. A voltage can be set on each bit line to allow or inhibit programming during each program voltage.

(105) The region **DL19a** has the example memory holes **710** and **711** along a line **712a** which is coincident with a bit line **BL0**. A number of bit lines extend above the memory holes and are connected to the memory holes as indicated by the “X” symbols. **BL0** is connected to a set of memory holes which includes the memory holes **711**, **715**, **717** and **719**. Another example bit line **BL1** is connected to a set of memory holes which includes the memory holes **710**, **714**, **716** and **718**. The metal-filled slits **701**, **702**, **703** and **704** from FIG. 7A are also depicted, as they extend vertically through the stack. The bit lines can be numbered in a sequence **BL0-BL23** across the **DL19** layer in the -x direction.

(106) Different subsets of bit lines are connected to cells in different rows. For example, **BL0**, **BL4**, **BL8**, **BL12**, **BL16** and **BL20** are connected to cells in a first row of cells at the right hand edge of each region. **BL2**, **BL6**, **BL10**, **BL14**, **BL18** and **BL22** are connected to cells in an adjacent row of cells, adjacent to the first row at the right hand edge. **BL3**, **BL7**, **BL11**, **BL15**, **BL19** and **BL23** are connected to cells in a first row of cells at the left hand edge of each region. **BL1**, **BL5**, **BL9**, **BL13**, **BL17** and **BL21** are connected to cells in an adjacent row of cells, adjacent to the first row at the left hand edge.

(107) FIG. 8A depicts example NAND strings in the sub-blocks **SBa-SBd** of FIG. 7A. The sub-blocks are consistent with the structure of FIG. 6B. The conductive layers in the stack are depicted for reference at the left hand side. Each sub-block includes multiple NAND strings, where one example NAND string is depicted. For example, **SBa** comprises an example NAND string **NS0_SBa**, **SBb** comprises an example NAND string **NS0_SBb**, **SBc** comprises an example NAND string **NS0_SBc**, and **SBd** comprises an example NAND string **NS0_SBd**.

(108) Additionally, **NS0_SBa** include SGS transistors **800** and **801**, dummy memory cells **802** and **803**, data memory cells **804**, **805**, **806**, **807**, **808**, **809**, **810**, **811**, **812**, **813** and **814**, dummy memory cells **815** and **816**, and SGD transistors **817** and **818**.

(109) **NS0_SBb** include SGS transistors **820** and **821**, dummy memory cells **822** and **823**, data memory cells **824**, **825**, **826**, **827**, **828**, **829**, **830**, **831**, **832**, **833** and **834**, dummy memory cells **835** and **836**, and SGD transistors **837** and **838**.

(110) **NS0_SBc** include SGS transistors **840** and **841**, dummy memory cells **842** and **843**, data memory cells **844**, **845**, **846**, **847**, **848**, **849**, **850**, **851**, **852**, **853** and **854**, dummy memory cells **855** and **856**, and SGD transistors **857** and **858**.

(111) **NS0_SBd** include SGS transistors **860** and **861**, dummy memory cells **862** and **863**, data memory cells **864**, **865**, **866**, **867**, **868**, **869**, **870**, **871**, **872**, **873** and **874**, dummy memory cells **875** and **876**, and SGD transistors **877** and **878**.

(112) At a given height in the block, a set of memory cells in each sub-block are at a common height. For example, one set of memory cells (including the memory cell **804**) is among a plurality of memory cells formed along tapered memory holes in a stack of alternating conductive and dielectric layers. The one set of memory cells is at a particular height **z0** in the stack. Another set of memory cells (including the memory cell **824**) connected to the one word line (**WLL0**) are also at the particular height. In another approach, the set of memory cells (e.g., including the memory cell **812**) connected to another word line (e.g., **WLL8**) are at another height (**z8**) in the stack.

(113) FIG. 8B depicts another example view of NAND strings in sub-blocks. The NAND strings includes **NS0_SBa**, **NS0_SBb**, **NS0_SBc** and **NS0_SBd**, which have 48 word lines, **WL0-WL47**, in this example. Each sub-block comprises a set of NAND strings which extend in the x direction and which have a common SGD line, e.g., **SGD0**, **SGD1**, **SGD2** or **SGD3**. In this simplified example, there is only one SGD transistor and one SGS transistor in each NAND string. The NAND strings **NS0_SBa**, **NS0_SBb**, **NS0_SBc** and **NS0_SBd** are in sub-blocks **SBa**, **SBb**, **SBc** and **SBd**, respectively. Further, example, groups of word lines **G0**, **G1** and **G2** are depicted.

(114) FIG. 8C generally illustrates a schematic view of three versions of staggered string architecture **101**, **103**, **105** for BiCS memory, e.g., NAND. With reference the string architecture **101**, the strings are shown in rows **107-0** through **107-7** in architecture **101**. Each row is shown

with four ends to the strings. A string may be connected to an adjacent string at an end (not visible beneath this view). A first group of rows **107-0** through **107-3** are shown on a left side of a dummy row **108**. A second group of rows **107-4** through **107-7** are shown on a right side of the dummy row **108**. The dummy row **108** separates the two groups of rows in the staggered eight rows. A source line **109** is positioned at an edge of the first group and is remote from the dummy row **108**. A source line **110** is positioned at an edge of the second group and is remote from the dummy row **108** and source line **109**.

(115) The staggered string architectures **103**, **105** for BiCS memory are similar to that of architecture **101** except additional groups are added. Architecture **103** is double the size of architecture **101** and includes sixteen rows of strings with each group of four rows separated by a dummy row. Architecture **105** is larger than both the architecture **101** and the architecture **103**. Architecture **105** includes twenty rows of strings with each group of four rows separated by a dummy row **108**.

(116) These architectures **101**, **103**, **105** can include a chip under array structure, e.g., the control circuitry is under the memory array that can include the groups of memory strings. With the chip under array structure, the strings may include a direct strap contact for the source line for read and erase operations.

(117) FIG. **12** depicts a waveform of an example programming operation. The horizontal axis depicts a program loop number and the vertical axis depicts control gate or word line voltage. Generally, a programming operation can involve applying a pulse train to a selected word line, where the pulse train includes multiple program loops or program-verify (PV) iterations. The program portion of the program-verify iteration comprises a program voltage, and the verify portion of the program-verify iteration comprises one or more verify voltages.

(118) For each program voltage, a square waveform is depicted for simplicity, although other shapes are possible such as a multilevel shape or a ramped shape. Further, Incremental Step Pulse Programming (ISPP) is used in this example, in which the program voltage steps up in each successive program loop. This example uses ISPP in a single programming stage in which the programming is completed. ISPP can also be used in each programming stage of a multi-stage operation.

(119) A pulse train typically includes program voltages which increase stepwise in amplitude in each program-verify iteration using a fixed or varying step size. A new pulse train can be applied in each programming stage of a multi-stage programming operation, starting at an initial V_{pgm} level and ending at a final V_{pgm} level which does not exceed a maximum allowed level. The initial V_{pgm} levels can be the same or different in different programming stages. The final V_{pgm} levels can also be the same or different in different programming stages. The step size can be the same or different in the different programming stages. In some cases, a smaller step size is used in a final programming stage to reduce V_{th} distribution widths.

(120) The pulse train **900** includes a series of program voltages **901**, **902**, **903**, **904**, **905**, **906**, **907**, **908**, **909**, **910**, **911**, **912**, **913**, **914** and **915** that are applied to a word line selected for programming, and an associated set of non-volatile memory cells. One, two or three verify voltages are provided after each program voltage as an example, based on the target data states which are being verified. 0 V may be applied to the selected word line between the program and verify voltages. For example, an A-state verify voltage of V_{vA} (e.g., waveform or programming signal **916**) may be applied after each of the first, second and third program voltages **901**, **902** and **903**, respectively. A- and B-state verify voltages of V_{vA} and V_{vB} (e.g., programming signal **917**) may be applied after each of the fourth, fifth and sixth program voltages **904**, **905** and **906**, respectively. A-, B- and C-state verify voltages of V_{vA} , V_{vB} and V_{vC} (e.g., programming signal **918**) may be applied after each of the seventh and eighth program voltages **907** and **908**, respectively. B- and C-state verify voltages of V_{vB} and V_{vC} (e.g., programming signal **919**) may be applied after each of the ninth, tenth and eleventh program voltages **909**, **910** and **911**, respectively. Finally, a C-state verify

voltage of VvC (e.g., programming signal **1020**) may be applied after each of the twelfth, thirteenth, fourteenth and fifteenth program voltages **912**, **913**, **914** and **915**, respectively.

(121) FIGS. **13A** and **13B** show threshold voltage (V_{th}) distributions of memory cells in an example two-stage programming operation. Specifically, the memory cells are initially in the erased state (bits **11**) as represented by the V_{th} distribution **1000** shown in FIG. **13A**. FIG. **13B** depicts V_{th} distributions of memory cells after a first programming stage and a second programming stage of the example two-stage programming operation with four data states. While two programming stages and four data states are shown, it should be appreciated that any number of programming stages may be utilized (e.g., three or four programming stages) and any number of data states are contemplated.

(122) In the example, the first programming stage causes the V_{th} of the A, B and C state cells to reach the V_{th} distributions **1002a**, **1004a** and **1006a**, using first verify voltages of VvAf, VvBf and VvCf, respectively. This first programming stage can be a rough programming which uses a relatively large step size, for instance, so that the V_{th} distributions **1002a**, **1004a** and **1006a** are relatively wide. The second programming stage may use a smaller step size and causes the V_{th} distributions **1002a**, **1004a** and **1006a** to transition to the final V_{th} distributions **1002**, **1004** and **1006** (e.g., narrower than V_{th} distributions **1002a**, **1004a** and **1006a**), using second verify voltages of VvA, VvB, and VvC, respectively. This two-stage programming operation can achieve relatively narrow V_{th} distributions. A small number of A, B and C state cells (e.g., smaller than a predetermined number of the plurality of memory cells) may have a V_{th} which is below VvA, VvB or VvC, respectively, due to a bit ignore criteria.

(123) FIGS. **14A** and **14B** provide V_{th} distributions of a memory device array. Conventionally, programmed state levels are higher than a V_{th} value of 0V because a V_{th} value below 0V cannot be sensed or verified. In FIG. **14A**, the V_{th} distribution without negative wordline (NWL) depicts higher states (which are further from neutral V_{th} than lower states) having large vertical data retention (DR) degradation. This issue is worse for the end-of-life (EOL) stage of the memory device, where the vertical DR degradation is much larger due to the wear-out of the tunneling oxide. NWL is a scheme used in reducing DR degradation. As depicted in FIG. **14B**, the V_{th} distribution with NWL depicts implementing NWL scheme, where the erase/program window and all states V_{th} can be shifted down, thereby allowing the higher states to be closer to the neutral V_{th} level. As a result, the vertical DR degradation may be improved.

(124) Due to aggressive scaling and cost reduction, there is new process improvement, such as smaller ON pitch, Y-size shrink, and MANOS improvement. After introducing NWL, many process changes can significantly impact the DR degradation for begin-of-life (BOL) stage of the product lifetime of the memory device. However, not implementing NWL would lose the benefit for the EOL stage for a memory device. To resolve this, embodiments described herein are directed to an adaptive NWL scheme which helps reduce DR degradation for the BOL stage but simultaneously maintaining NWL benefit for the EOL stage.

(125) Conventionally, as depicted in FIG. **15A**, a NWL scheme includes application of a static negative wordline voltage (NWLTV). For example, a static NWLTV is applied at the BOL, MOL middle-of-life (MOL), and EOL stages of the memory device product lifetime. Embodiments described herein are directed to a scheme of applying adaptive NWLTV based on a stage of the memory device product lifetime. For example, as depicted in FIG. **15B**, a different NWLTV may be set at different stages of the memory device product lifetime. For example, NWLTV may not be implemented for BOL stage (e.g., NWLTV=0 is applied for 0-500 cycles), NWLTV may be set to a first value for MOL stage (e.g., NWLTV=1.1V is applied for 500-1000 cycles), and NWLTV may be set to a second value, which is higher than the first value for EOL stage (e.g., NWLTV=1.2V is applied for 1000-3000 cycles). This provides a benefit of a gain of DR V_{th} margin at BOL stage of a memory device, while maintaining the NWL benefit at EOL stage of the memory device without sacrificing performance.

(126) In some embodiments, a single parameter may be used to adaptively adjust any device parameters related to NWL based on the stages of the memory device product lifetime. For example, a single parameter may be used to adjust the following parameters for various operations: an erase voltage (e.g., VERA, VERAMAX, etc.), erase verify (e.g., VCG_ERV), program voltage (e.g., VPGMU, VPGMMAX), program unselect-wordline bias (e.g., VREAD, VPASS, etc.), read unselect-wordline bias (VREAD), and verify and read select wordline (e.g., VCG for all states). The benefits of an adaptive NWL scheme provides DR degradation benefit of EOL blocks without impacting BOL blocks.

(127) In determining NWL related settings for different stages of a memory device product lifetime, an age-NWL setting correlation may be established. In some embodiments, age of a memory device may be detected by a smart-verify operation. After E/P cycling and wear-out of memory cells, a program speed will be faster. For example, the faster program speeds cause lower program voltage (e.g., VPGMU) to program state A and can be detected by smart verify in A-state sensing. In accordance with embodiments described herein, age of the memory device detected by smart-verify will enable a mapping between smart verify VPGMU and age (e.g., E/P cycles) and the NWL setting (e.g., NWLV). For example, a smart-verify VPGMU of 13.5V may map to cycles of 0-999 and an NWLV of 0V, a smart-verify VPGMU of 13.4V may map to cycles of 1000-1999 and an NWLV of 1.1V, and a smart-verify VPGMU of 13.3V may map to cycles over 1999 and an NWLV of 2.2V.

(128) In some embodiments, a NWL setting may be stored in a test mode matrix (TMM) table. The TMM table with a parameter adjustment may be stored in firmware, while age of the memory device is also recorded by firmware. With the TMM table, parameters may be adjusted according to the NWL setting corresponding to the age of the memory device stored in the TMM table. The NWL setting can also be stored in TMM table for adjusting NWL relative parameters based on the age.

(129) In some embodiments, an adaptive NWLV scheme may be implemented for different hybrid xLC mode. As described, SLC can refer to a cell programmed to a targeted one of two different data states and configured to store a single data unit (e.g., one bit). Some memory cells can be programmed to a targeted one of more than two different data states such that they are configured to store more than a single data unit (e.g., 2 bits, 3 bits, 4 bits, etc.). Such cells may be referred to as multi state memory cells or extra level cells (XLCs). XLCs can provide higher density memories without increasing the number of memory cells since each cell can represent more than one data unit. A storage region may have a SLC configuration, a MLC configuration, a TLC configuration, or a QLC configuration. Furthermore, some memory devices may have more than one storage region type and is not limited to having a uniform storage region type (i.e., having both SLC and TLC).

(130) Conventionally, as shown in FIG. 16A, a static NWLV scheme is implemented regardless of xLC mode of the memory device. A static NWLV may not be optimal for every xLC mode. For example, as shown in FIG. 16A, what is optimal for SLC may not be optimal for TLC or QLC. TLC and QLC will have larger vertical DR than SLC when implementing a static NWLV scheme. Multiple data storage modes can be used in hybrid xLC mode (e.g., SLC for high performance and then switching to QLC for higher capacity when more data is stored). However for different xLC modes, the high state level to neutral V_{th} voltage drop is different. The same NWLV will not be optimal for every mode. As depicted in FIG. 16B, using an adaptive NWLV for different xLC modes so that NWLV is optimal for each xLC mode will minimize vertical DR in hybrid xLC modes (e.g., SLC-TLC hybrid, SLC-QLC hybrid, etc.).

(131) There is an optimal NWLV that offers best DR degradation (mostly in BOL blocks). When the NWLV is too high or too low the DR degradation will be worse. The optimal NWLV can vary with process and also vary between QLC and TLC. For example, for QLC the highest state level is higher than TLC, so it will need higher NWL to reduce DR, and the optimal NWLV will be higher.

(132) Furthermore, word lines and strings are in different physical environments, and thus larger variations are seen when scaling up along vertical (more word lines) and lateral (more memory holes) directions. Inner memory holes and bottom word lines usually suffer worse fluorine-degas, word line metal filling, and barrier film coverage issues. During program or erase phase, all memory rows in one string will be programmed or erased at the same time. Due to the physical variations, the device behavior of cells, such as erase saturation and program-erase window, will be different. Moreover, severe erase tail up-shift is observed during P/E cycling, which results in erase status fail eventually.

(133) To address erase-program window word line, string, and EP-cycle variations, embodiments described herein are directed to erase-program window oriented cell source voltage (e.g., V_{celsrc}) tuning for word line, string, and EP-cycle variation compensation. For example, an erase upper tail be detected and an adaptive V_{celsrc} may be implemented with DS/device trim at different word line, string, and EP-cycle levels.

(134) In some embodiments, as shown in FIG. 17, an erase upper tail is detected. For example, an erase verify voltage (V_{CG_ERV}) scan may be performed to detect an erase upper tail after a first erase pulse. With reference to FIGS. 1A and 1B, control circuitry 110 may determine an upper tail of a threshold voltage distribution of the first grouping of non-volatile storage elements (e.g., storage device 126a). More specifically, control circuitry 110 may determine a reference point on the threshold distribution after the first erase pulse. The reference point as referred to herein is an “upper tail V_{th} ,” as the reference point is typically on the very upper end of the V_{th} distribution. In some embodiments, an erase verify voltage scan may be used to determine a reference point on the threshold distribution.

(135) Further, in FIG. 17, at the judge with criteria step, a reference value for optimal erase upper tail level is judged. At this step, after the erase upper tail is detected, the erase upper tail is compared with the reference value. If it is higher or lower than the reference value beyond the tolerance, V_{celsrc} is adjusted.

(136) As further depicted in FIG. 17, V_{celsrc} adjust amount is determined based on the difference between the data of a wordline and a pre-determined reference value. This will achieve a uniform erase-program window, and the erase-verify level will shift according to the V_{celsrc} adjustment, resulting in less and later erase status fail. The memory device life or endurance will be improved, as the erase status failures will be reduce and postponed.

(137) To help further explore the above in further detail, FIG. 18 will now be described. FIG. 18 depicts a method of implementing an adaptive NWLV scheme, in accordance with embodiments described herein. In some embodiments, method 1800 may be implemented by a controller, control circuitry, a processor, and/or the like, as described elsewhere herein. As shown in FIG. 18, method 1800 starts at step 1802. In step 1802, a negative word line setting, for implementing during performance of a first operation, is determined based on a stage of a product lifetime of the non-volatile memory. For example, with reference to FIGS. 1A and 1B, control circuitry 110 may determine, based on a stage of a product lifetime of the non-volatile memory a negative word line setting for implementing during performance of a first operation.

(138) In FIG. 18, at step 1804, the first operation is performed, where the first operation includes adjusting, based on the negative word line setting, a negative word line relative parameter. As described, with continued reference to FIGS. 1A and 1B, control circuitry 110 may perform the first operation, the first operation including adjusting, based on the negative word line setting, a negative word line relative parameter.

(139) In FIG. 18, at step 1806, another negative word line setting, for implementing during performance of a second operation, is determined based on another stage of the product lifetime of the non-volatile memory. As described, with continued reference to FIGS. 1A and 1B, control circuitry 110 may determine, based on another stage of the product lifetime of the non-volatile memory, another negative word line setting for implementing during performance of a second

operation.

(140) In FIG. 18, at step 1808, the second operation is performed, where the second operation includes adjusting, based on the other negative word line setting, another negative word line relative parameter. As described, with continued reference to FIGS. 1A and 1B, control circuitry 110 may perform the second operation, the second operation including adjusting, based on the other negative word line setting, another negative word line relative parameter.

(141) The foregoing detailed description of the invention has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise form disclosed. Many modifications and variations are possible in light of the above teachings. The described embodiments were chosen in order to best explain the principles of the invention and its practical application, to thereby enable others skilled in the art to best utilize the invention in various embodiments and with various modifications as are suited to the particular use contemplated. It is intended that the scope of the invention be defined by the claims appended hereto.

Claims

1. A storage device, comprising: a non-volatile memory including control circuitry and an array of memory cells formed using a set of word lines and a set of bit lines; and a controller, coupled to the non-volatile memory, configured to: determine, based on a first stage of a product lifetime of the non-volatile memory, a first negative word line setting to be implemented during a first operation; perform, in the first stage, the first operation, the first operation including applying, based on the first negative word line setting, a first negative word line voltage; determine, based on a second stage of the product lifetime of the non-volatile memory subsequent to the first stage, a second negative word line setting to be implemented during a second operation; and perform, in the second stage, the second operation, the second operation including applying, based on the second negative word line setting, a second negative word line voltage, wherein the second negative word line voltage is more negative than the first negative word line voltage, wherein the first negative word line setting varies based on a selected multiple data storage mode.
2. The storage device of claim 1, wherein the controller is further configured to: determine the first stage of the product lifetime of the non-volatile memory by detecting erase/program cycles of the non-volatile memory.
3. The storage device of claim 1, wherein the controller is further configured to: determine the second stage of the product lifetime of the non-volatile memory by detecting erase/program cycles of the non-volatile memory.
4. The storage device of claim 1, wherein the controller is further configured to: perform a smart verify operation to determine a program voltage; and determine, based on the program voltage, the first negative word line setting.
5. The storage device of claim 1, wherein the controller is further configured to: perform a smart verify operation to determine a program voltage; and determine, based on the program voltage, the second negative word line setting.
6. The storage device of claim 1, wherein the controller is further configured to: determine an upper tail of a threshold voltage distribution; and apply, based on the upper tail, an adaptive cell source voltage on different word line, string, and erase/program cycle variations.
7. A method of operating a non-volatile semiconductor memory device, the method comprising: determining, based on a first stage of a product lifetime of a non-volatile memory, a first negative word line setting to be implemented during a first operation; performing, in the first stage, the first operation, the first operation including applying, based on the first negative word line setting, a first negative word line voltage; determining, based on a second stage of the product lifetime of the non-volatile memory subsequent to the first stage, a second negative word line setting to be

- implemented during a second operation; and performing, in the second stage, the second operation, the second operation including applying, based on the second negative word line setting, a second negative word line voltage, wherein the second negative word line voltage is more negative than the first negative word line voltage, wherein the first negative word line setting varies based on a selected multiple data storage mode.
8. The method of claim 7, further comprising: determining the first stage of the product lifetime of the non-volatile memory by detecting erase/program cycles of the non-volatile memory.
9. The method of claim 7, further comprising: determining the second stage of the product lifetime of the non-volatile memory by detecting erase/program cycles of the non-volatile memory.
10. The method of claim 7, further comprising: performing a smart verify operation to determine a program voltage; and determining, based on the program voltage, the first negative word line setting.
11. The method of claim 7, further comprising: performing a smart verify operation to determine a program voltage; and determining, based on the program voltage, the second negative word line setting.
12. The method of claim 7, further comprising: determining, based on a multiple data storage mode of the non-volatile memory, the first negative word line setting to be implemented during the first operation.
13. The method of claim 7, further comprising: determining an upper tail of a threshold voltage distribution; and applying, based on the upper tail, an adaptive cell source voltage on different word line, string, and erase/program cycle variations.
14. An apparatus comprising: means for determining, based on a first stage of a product lifetime of a non-volatile memory, a first negative word line setting to be implemented during performance of a first operation; means for performing, in the first stage, the first operation, the first operation including applying, based on the first negative word line setting, a first negative word line voltage; means for determining, based on a second stage of the product lifetime of the non-volatile memory subsequent to the first stage, a second negative word line setting to be implemented during a second operation; and means for performing, in the second stage, the second operation, the second operation including applying, based on the second negative word line setting, a second negative word line voltage, wherein the second negative word line voltage is more negative than the first negative word line voltage, wherein the first negative word line setting varies based on a selected multiple data storage mode.
15. The apparatus of claim 14, further comprising: means for determining the first stage of the product lifetime of the non-volatile memory by detecting erase/program cycles of the non-volatile memory.
16. The apparatus of claim 14, further comprising: means for determining the second stage of the product lifetime of the non-volatile memory by detecting erase/program cycles of the non-volatile memory.
17. The apparatus of claim 14, further comprising: means for performing a smart verify operation to determine a program voltage; and means for determining, based on the program voltage, the first negative word line setting.
18. The apparatus of claim 14, further comprising: means for performing a smart verify operation to determine a program voltage; and means for determining, based on the program voltage, the second negative word line setting.
19. The apparatus of claim 14, further comprising: means for determining, based on a multiple data storage mode of the non-volatile memory, the first negative word line setting to be implemented during the first operation.
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